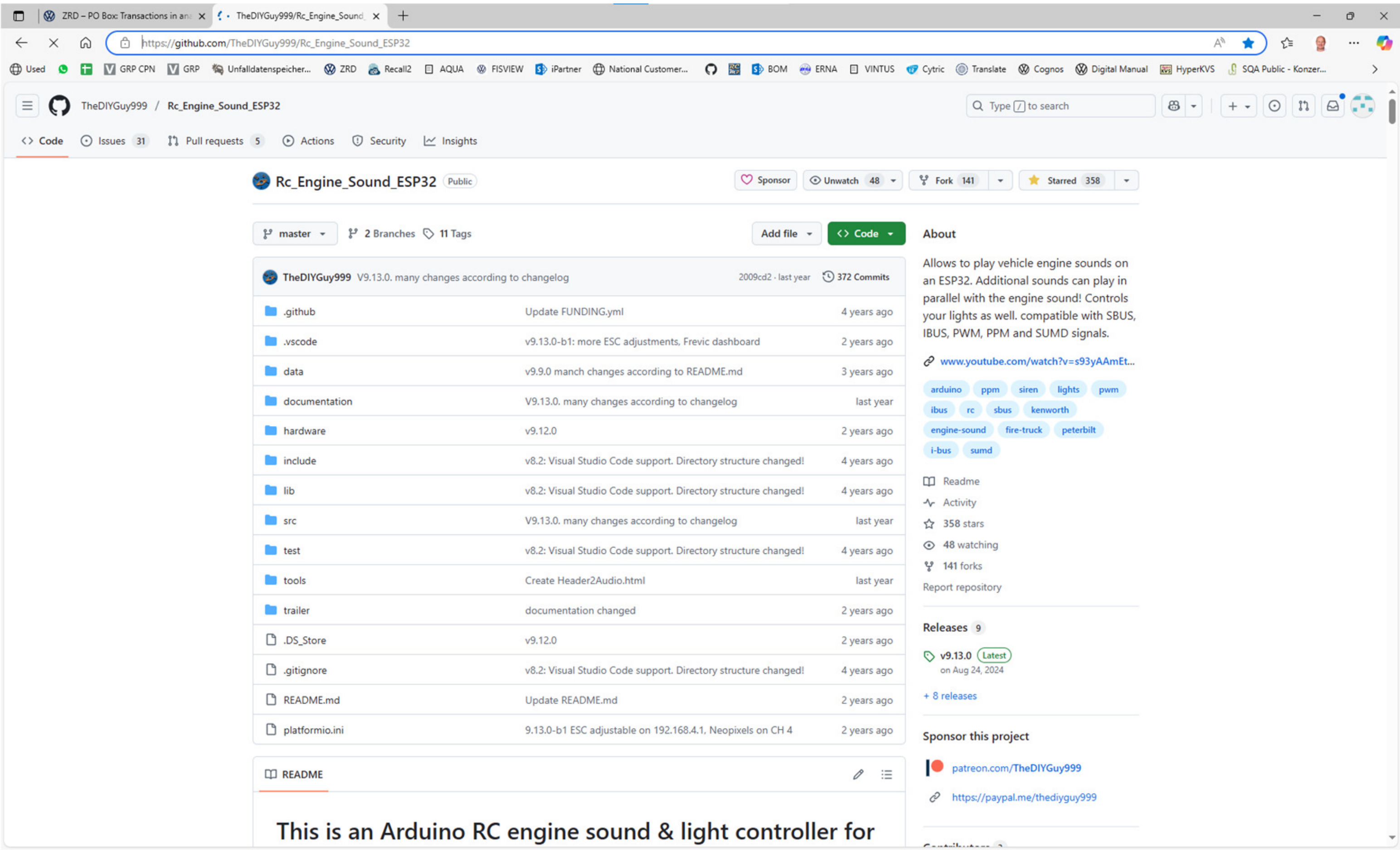
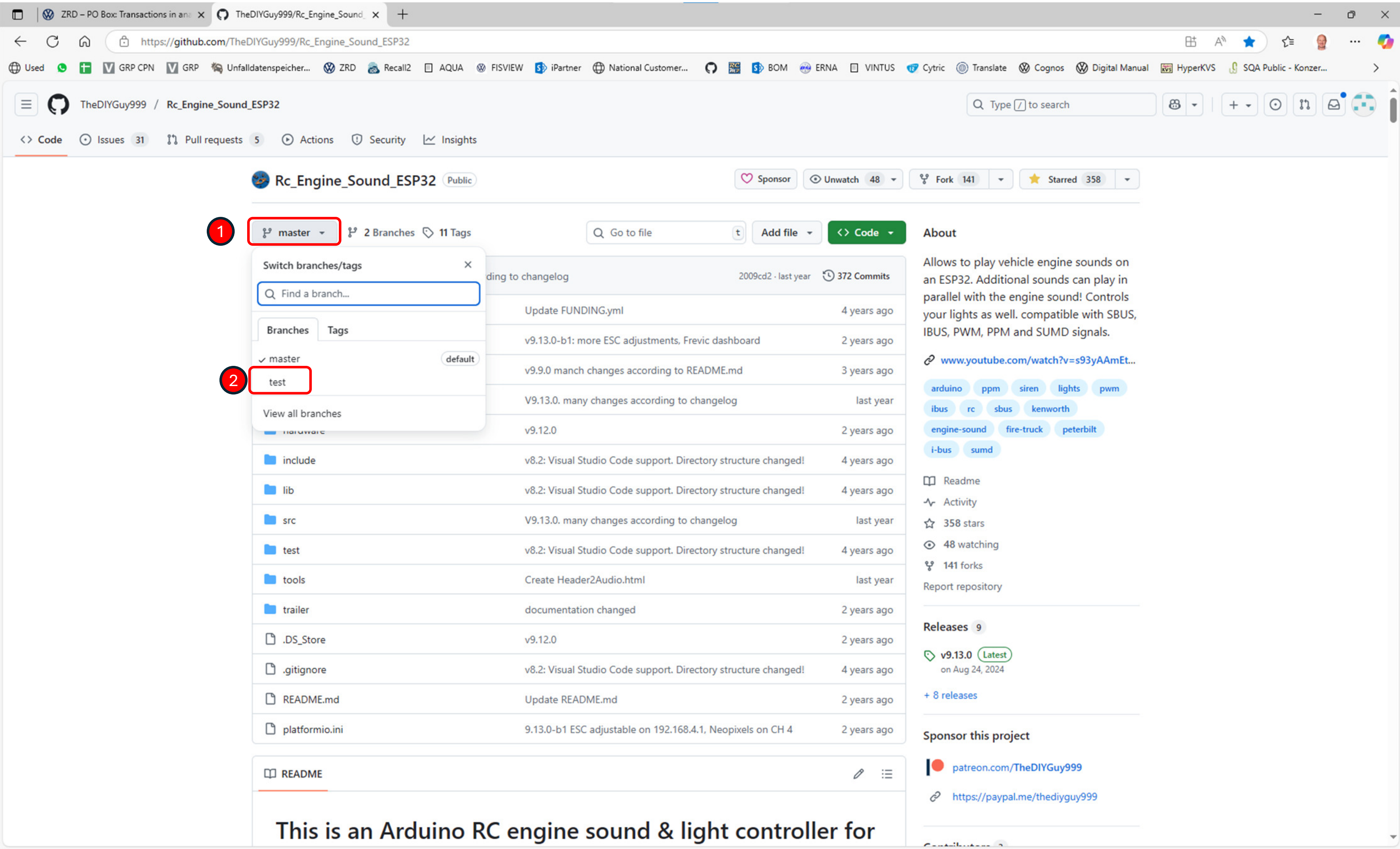
[TheDIYGuy999/Rc_Engine_Sound_ESP32: Allows to play vehicle engine sounds on an ESP32. Additional sounds can play in parallel with the engine sound! Controls your lights as well. compatible with SBUS, IBUS, PWM, PPM and SUMD signals.]




TheDIYGuy999/Rc_Engine_Sound_ESP32
https://github.com/TheDIYGuy999/Rc_Engine_Sound_ESP32
Code
Issues 31
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Rc_Engine_Sound_ESP32
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2009cd2 · last year
372 Commits
Update FUNDING.yml
v9.13.0-b1: more ESC adjustments, Frevic dashboard
v9.9.0 manch changes according to README.md
V9.13.0. many changes according to changelog
include
lib
src
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tools
trailer
.DS_Store
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README.md
platformio.ini
v8.2: Visual Studio Code support. Directory structure changed!
Create Header2Audio.html
documentation changed
v9.12.0
Update README.md
9.13.0-b1 ESC adjustable on 192.168.4.1, Neopixels on CH 4
About
Allows to play vehicle engine sounds on an ESP32. Additional sounds can play in parallel with the engine sound! Controls your lights as well. compatible with SBUS, IBUS, PWM, PPM and SUMD signals.
www.youtube.com/watch?v=s93yAAmEt...
arduino
ppm
siren
lights
pwm
ibus
rc
sbus
kenworth
engine-sound
fire-truck
peterbilt
i-bus
sumd
Readme
Activity
358 stars
48 watching
141 forks
Report repository
Releases 9
v9.13.0
Latest
on Aug 24, 2024
+ 8 releases
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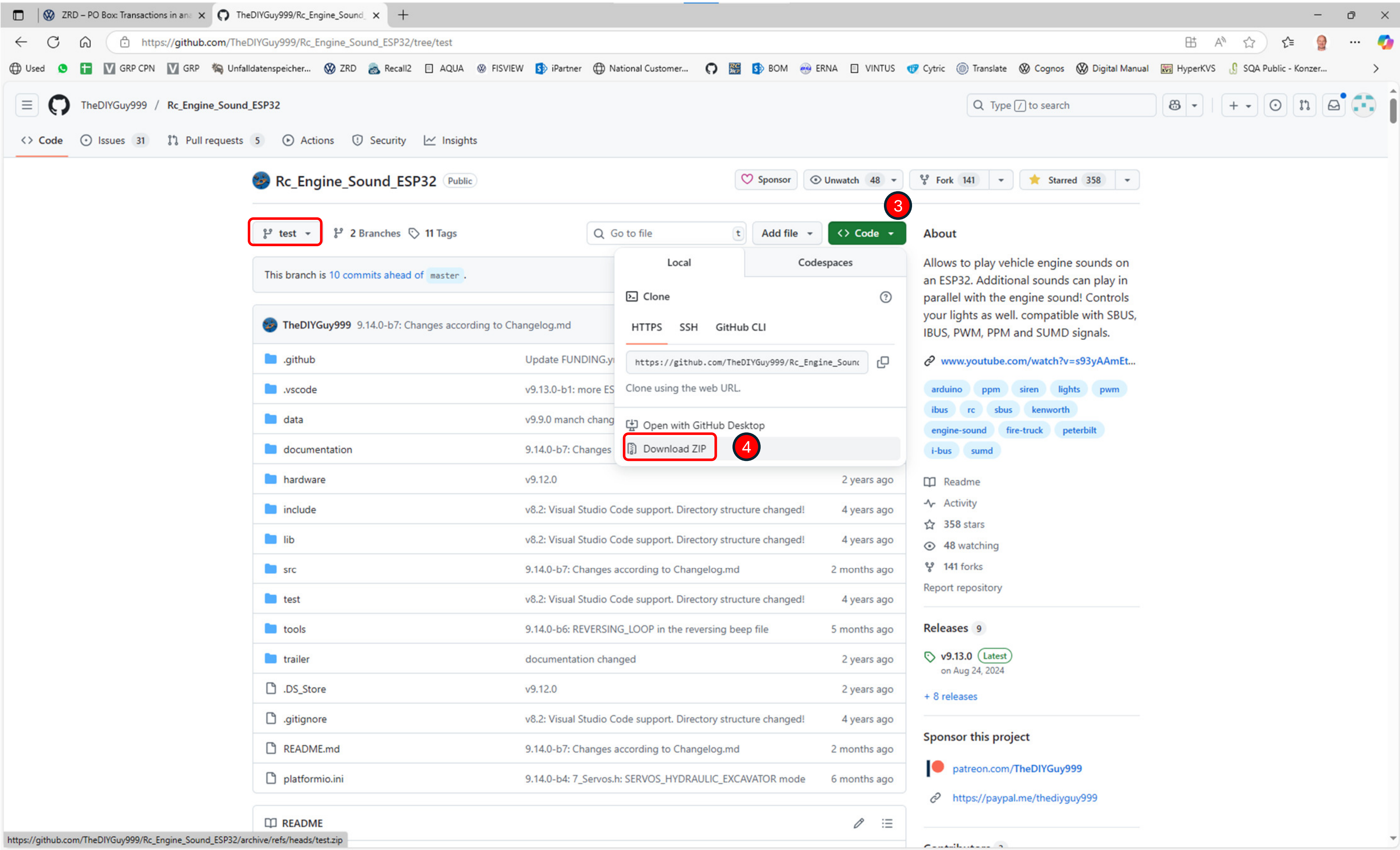
README
This is an Arduino RC engine sound & light controller for

INTERNAL


ZRD – PO Box: Transactions in an
TheDIYGuy999/Rc_Engine_Sound
https://github.com/TheDIYGuy999/Rc_Engine_Sound_ESP32/tree/test
TheDIYGuy999 / Rc_Engine_Sound_ESP32
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Unwatch 48
Fork 141
Starred 358
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2 Branches
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This branch is 10 commits ahead of master.
TheDIYGuy999 9.14.0-b7: Changes according to Changelog.md
Local
Codespaces
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SSH
GitHub CLI
https://github.com/TheDIYGuy999/Rc_Engine_Sounc
Clone using the web URL.
Open with GitHub Desktop
Download ZIP
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Update FUNDING.y
v9.13.0-b1: more ES
v9.9.0 manch chang
9.14.0-b7: Changes
v9.12.0
v8.2: Visual Studio Code support. Directory structure changed!
9.14.0-b7: Changes according to Changelog.md
9.14.0-b6: REVERSING_LOOP in the reversing beep file
documentation changed
9.14.0-b4: 7_Servos.h: SERVOS_HYDRAULIC_EXCAVATOR mode
2 years ago
4 years ago
2 months ago
5 months ago
6 months ago
README
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Allows to play vehicle engine sounds on an ESP32. Additional sounds can play in parallel with the engine sound! Controls your lights as well. compatible with SBUS, IBUS, PWM, PPM and SUMD signals.
www.youtube.com/watch?v=s93yAAmEt...
arduino
ppm
siren
lights
pwm
ibus
rc
sbus
kenworth
engine-sound
fire-truck
peterbilt
i-bus
sumd
Readme
Activity
358 stars
48 watching
141 forks
Report repository
Releases 9
v9.13.0
Latest
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+ 8 releases
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https://github.com/TheDIYGuy999/Rc_Engine_Sound_ESP32/archive/refs/heads/test.zip

MASTER

Does not contain Pingnon

Changelog

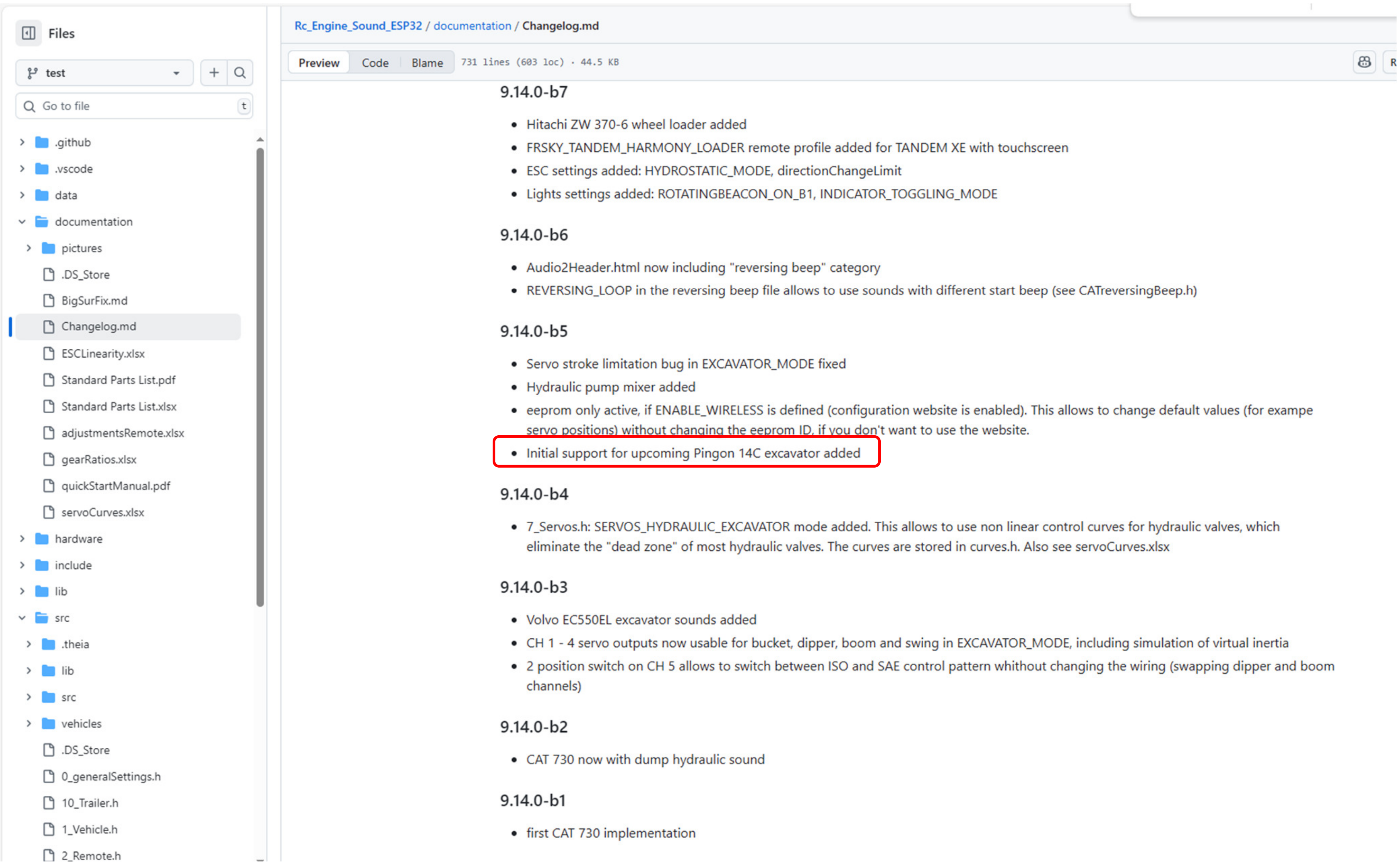
Todo:

- Remove in vehicle files: XENON_LIGHTS, LED_INDICATORS, INDICATOR_DIR, doubleFlashBlueLight
- Add to EEPROM & select box: DOUBLE_CLUTCH and other transmission options

9.13.0-b7:

- Defines removed: HAZARDS_WHILE_5TH_WHEEL_UNLOCKED
- Variables added (adjustable on 192.168.4.1): hazardsWhile5thWheelUnlocked
- Neopixel animation mode selectable in dropdown menu on 192.168.4.1
- Defines removed: NO_CABLIGHTS, NO_FOGLIGHTS, XENON_LIGHTS, FLICKERING_WHILE_CRANKING, SEPARATE_FULL_BEAM
- Variables removed: INDICATOR_DIR
- Variables added (adjustable on 192.168.4.1): noCabLights, noFogLights, xenonLights, flickeringWileCranking, swap_L_R_indicators, separateFullBeam, flashingBlueLight
- First light settings available on 192.168.4.1
- Website with collapsible sections to keep it more organized
- Servo positions adjustable on 192.168.4.1
- Improved website look
- Scania dashboard splash logo by Frikkiebester
- Dashboard requires less processing power, no engine rpm lagging anymore, if Frevic dashboard is used
- More adjustable ESC parameters on 192.168.4.1
- Dashboard improved & optional dashboard from Frevic implemented. Select it in "dashboard.h" by uncommenting "#define FREVIC_DASHBOARD"
- New parameter "#define NEOPIXEL_ON_CH4" in "6_Lights.h" allows to connect the NEOPIXELS to servo CH 4 (BUS mode only, no 5th wheel servo in this case)
- ESC parameters are adjustable via 192.168.4.1
- in "#define WEMOS_D1_MINI_ESP32" mode only: Experimental Serial command interface for ESC parameters

TEST



Files

test

Go to file

- .github
- .vscode
- data
- documentation
  - pictures
  - .DS_Store
  - BigSurFix.md
  - Changelog.md
  - ESCLinearity.xlsx
  - Standard Parts List.pdf
  - Standard Parts List.xlsx
  - adjustmentsRemote.xlsx
  - gearRatios.xlsx
  - quickStartManual.pdf
  - servoCurves.xlsx
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- include
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  - lib
  - src
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  - 10_Trailer.h
  - 1_Vehicle.h
  - 2_Remote.h

Rc_Engine_Sound_ESP32 / documentation / Changelog.md

Preview Code Blame 731 lines (603 loc) · 44.5 KB

9.14.0-b7

- Hitachi ZW 370-6 wheel loader added
- FRSKY_TANDEM_HARMONY_LOADER remote profile added for TANDEM XE with touchscreen
- ESC settings added: HYDROSTATIC_MODE, directionChangeLimit
- Lights settings added: ROTATINGBEACON_ON_B1, INDICATOR_TOGGLING_MODE

9.14.0-b6

- Audio2Header.html now including "reversing beep" category
- REVERSING_LOOP in the reversing beep file allows to use sounds with different start beep (see CATreversingBeep.h)

9.14.0-b5

- Servo stroke limitation bug in EXCAVATOR_MODE fixed
- Hydraulic pump mixer added
- eeprom only active, if ENABLE_WIRELESS is defined (configuration website is enabled). This allows to change default values (for exampe servo positions) without changing the eeprom ID, if you don't want to use the website.
- Initial support for upcoming Pingon 14C excavator added

9.14.0-b4

- 7_Servos.h: SERVOS_HYDRAULIC_EXCAVATOR mode added. This allows to use non linear control curves for hydraulic valves, which eliminate the "dead zone" of most hydraulic valves. The curves are stored in curves.h. Also see servoCurves.xlsx

9.14.0-b3

- Volvo EC550EL excavator sounds added
- CH 1 - 4 servo outputs now usable for bucket, dipper, boom and swing in EXCAVATOR_MODE, including simulation of virtual inertia
- 2 position switch on CH 5 allows to switch between ISO and SAE control pattern whithout changing the wiring (swapping dipper and boom channels)

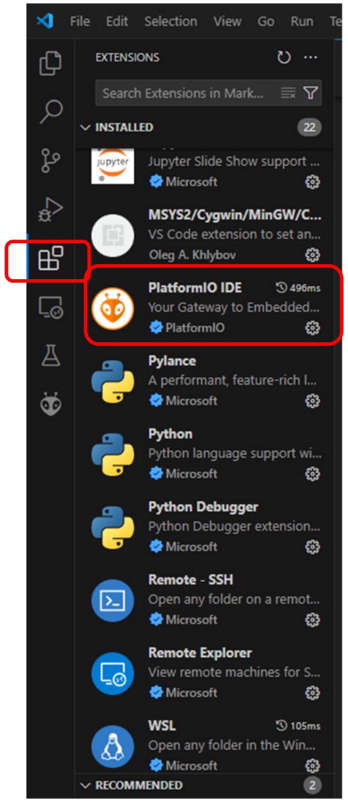
9.14.0-b2

- CAT 730 now with dump hydraulic sound

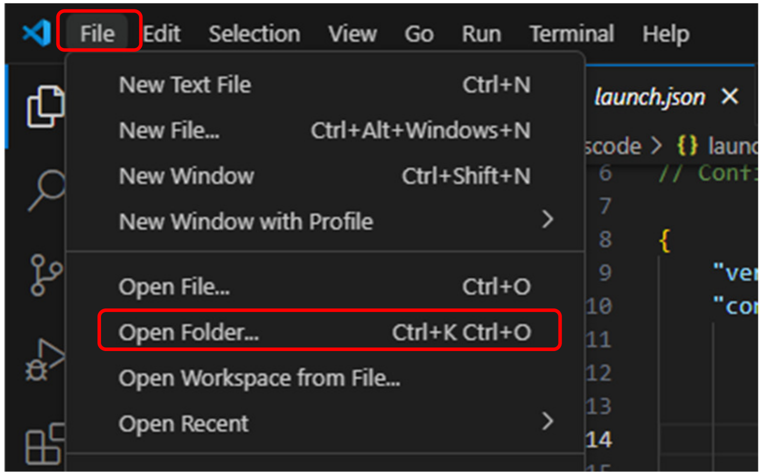
9.14.0-b1

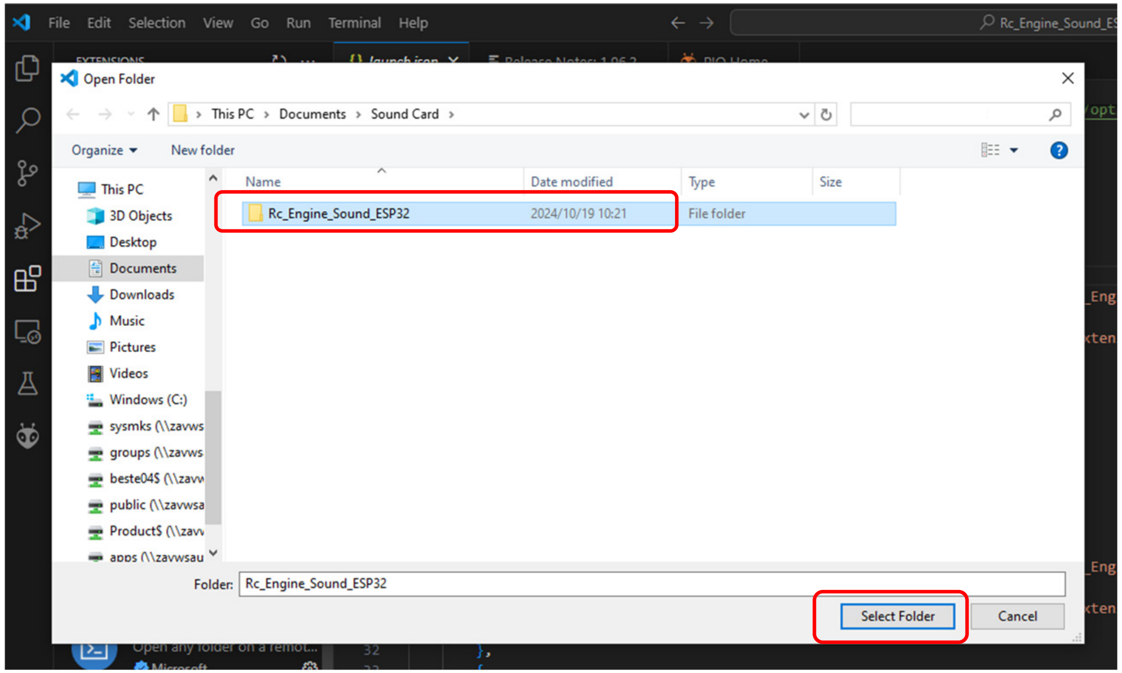
- first CAT 730 implementation

1. Install VSCode for you PC (Download Visual Studio Code - Mac, Linux, Windows)
2. Accept and install ALL add-inns that VSCode requires.
3. Restart VSCode. Once open click on icon below and ensure PlatformIO IDE is installed

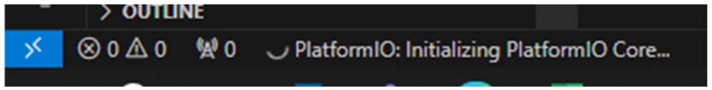


4. Open the downloaded folder from GITHUB

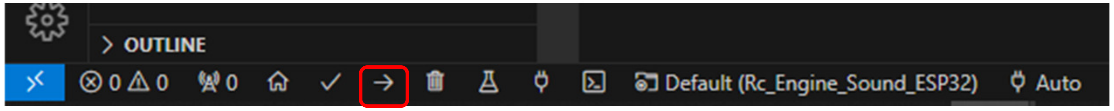




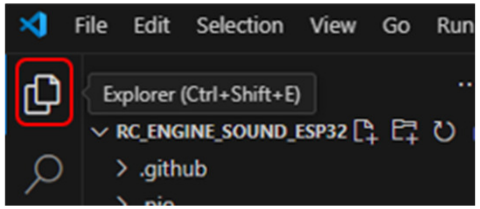
5. Allow the program to do what is required, accept and install any suggestions.
6. The first time it will run a bit longer to initialize everything

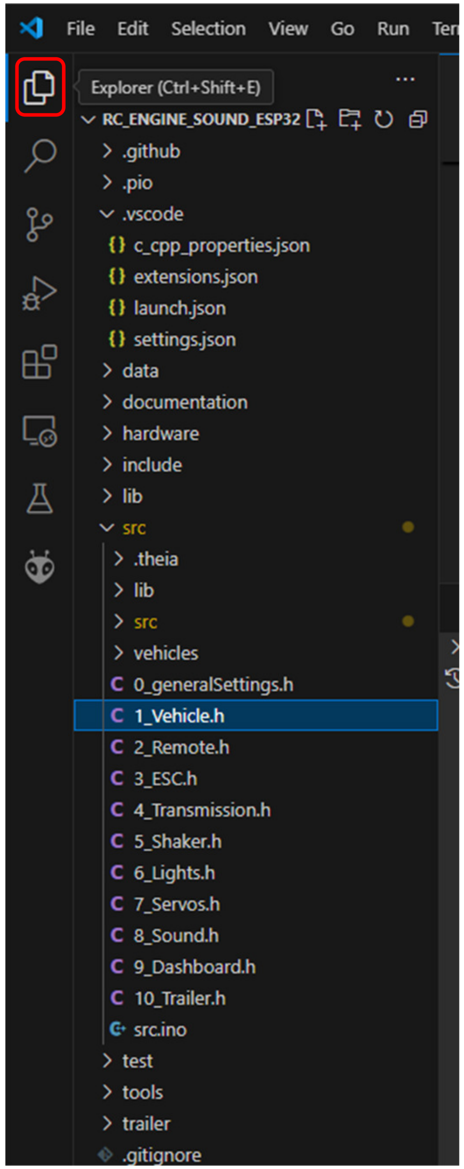


7. Once done you should have a bar like this at the bottom. The arrow is the upload button to the ESP32.



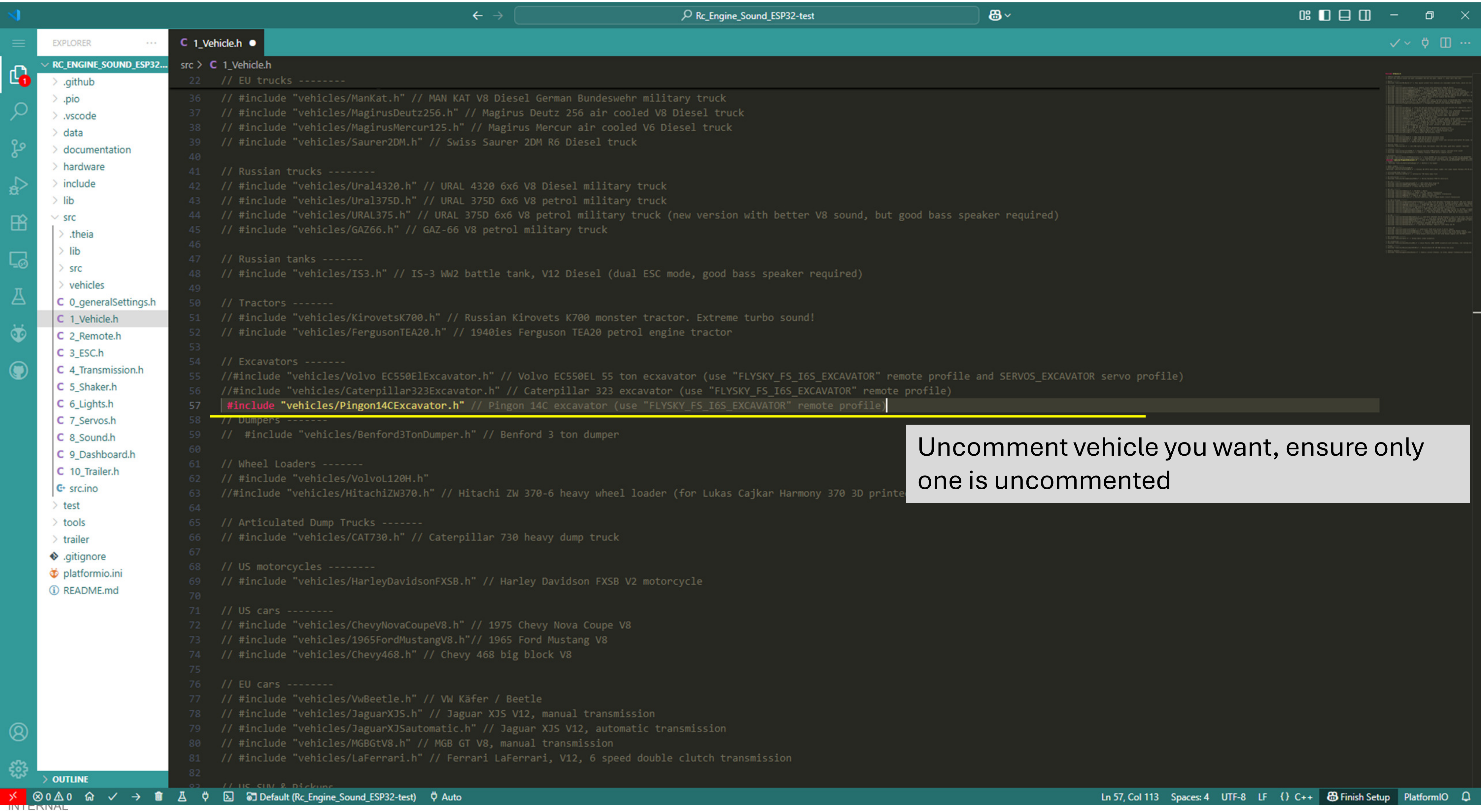
8. One more thing that you might need to install if you get any Squiggles errors is GIT. https://git-scm.com/downloads/win
9. To see all the folders you can click on icon.





To use Trailer Card, uncomment (remove //)
#define WEMOS_D1_MINI_ESP32
To use WiFi Communications, uncomment
//#define ENABLE_WIRELESS
src > 0_generalSettings.h
#include <Arduino.h>
#include <WiFi.h>
/* GENERAL SETTINGS
* Have a look at the serial monitor in order to check the current settings of your vehicle!
* The required login informations for the browser based configuration via 192.186.4.1 can also be found there.
// Hardware settings
#define WEMOS_D1_MINI_ESP32 // Software is running on ESP32 RC Trailer controller, if defined (Headlights on GPIO 22 instead of 3, no cab lights support)
// Wireless settings
#define ENABLE_WIRELESS // Define this, if you want to use an ESP-Now wireless trailer or the WiFi configuration via 192.168.4.1
wifi_power_t cpType = WIFI_POWER_7dBm; // Only use values from above!

INTERNAL


Rc_Engine_Sound_ESP32-test
EXPLORER
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3_ESC.h
4_Transmission.h
5_Shaker.h
6_Lights.h
7_Servos.h
8_Sound.h
9_Dashboard.h
10_Trailer.h
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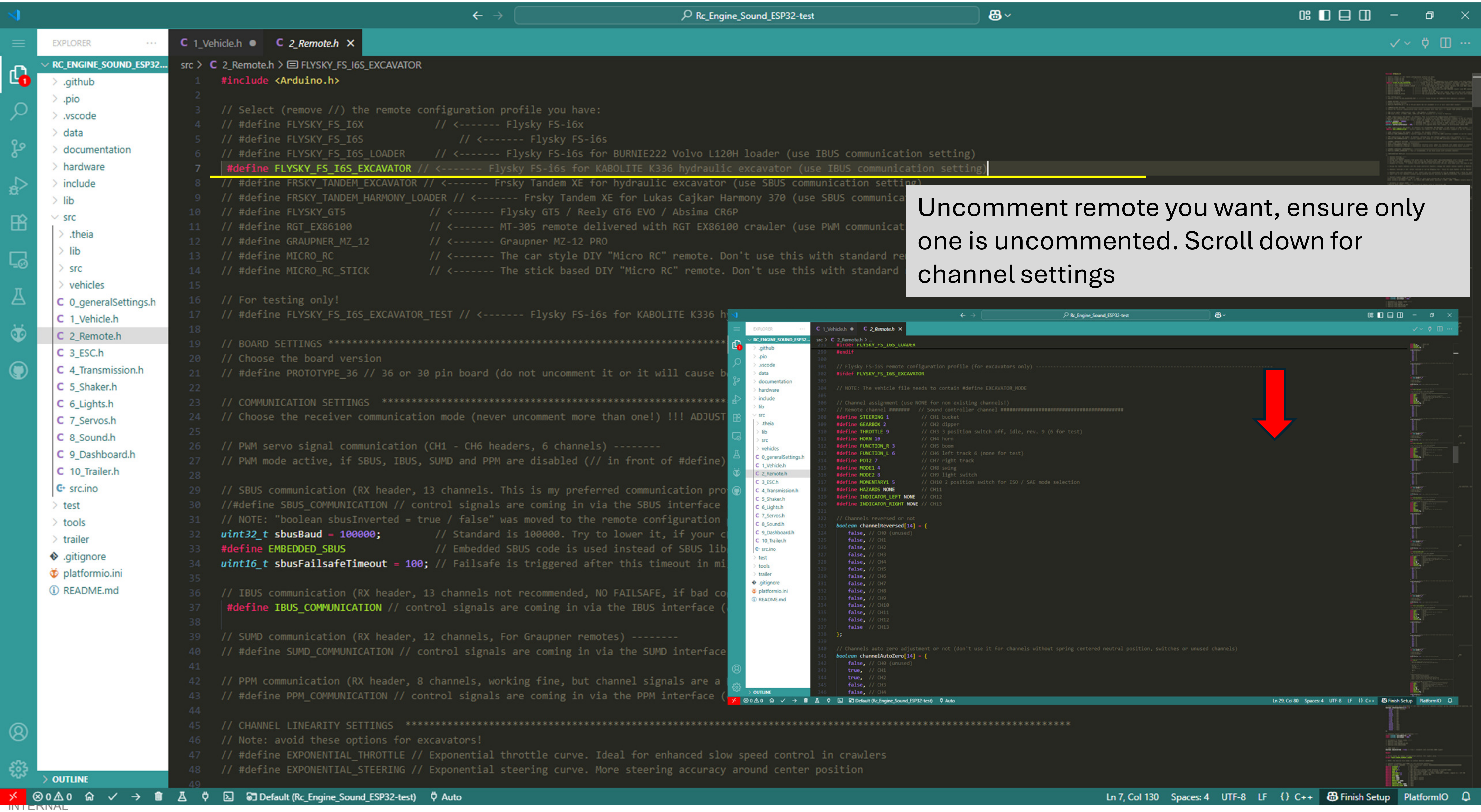
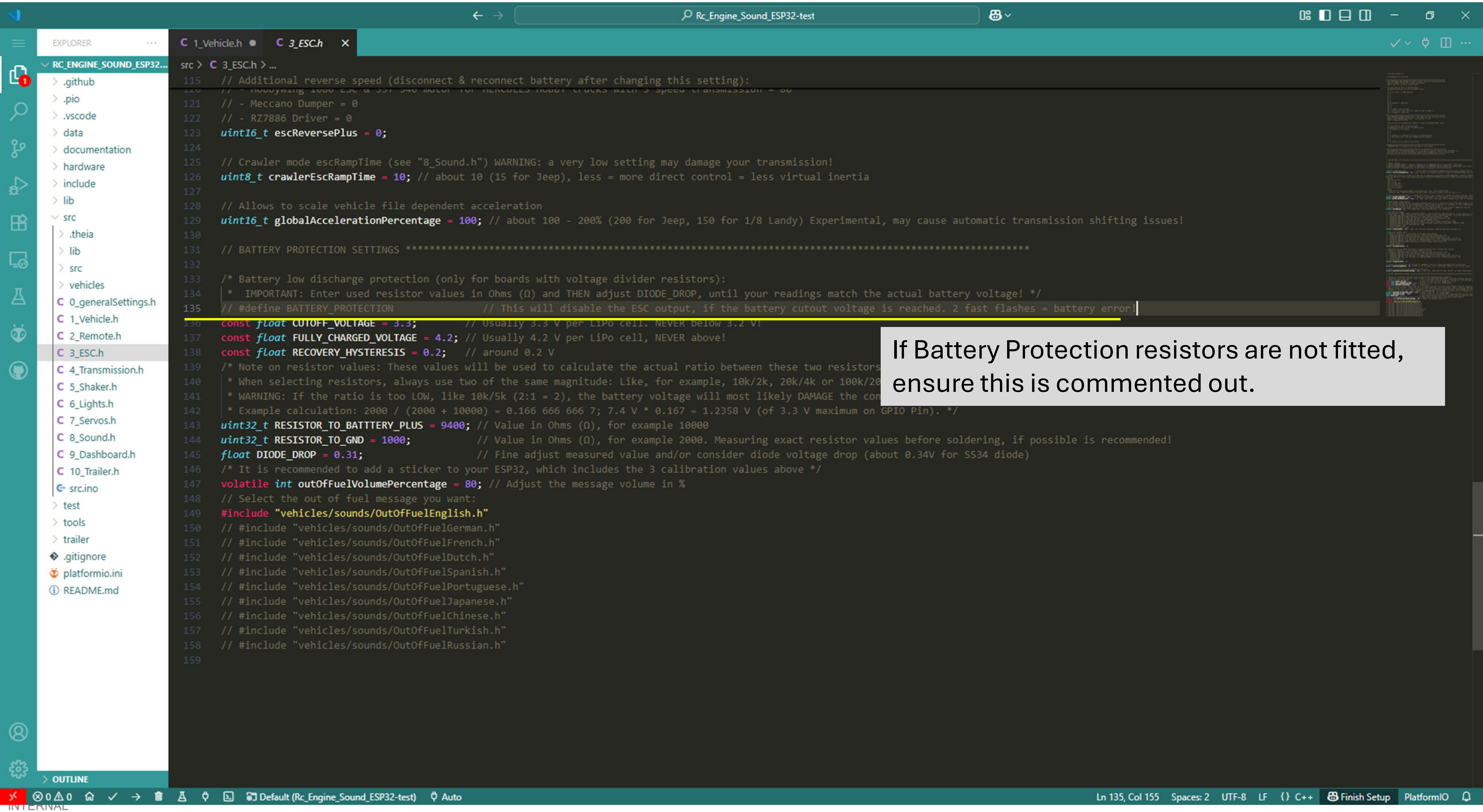
platformio.ini
README.md
OUTLINE
1_Vehicle.h
src > 1_Vehicle.h
// EU trucks --------
// #include "vehicles/ManKat.h" // MAN KAT V8 Diesel German Bundeswehr military truck
// #include "vehicles/MagirusDeutz256.h" // Magirus Deutz 256 air cooled V8 Diesel truck
// #include "vehicles/MagirusMercur125.h" // Magirus Mercur air cooled V6 Diesel truck
// #include "vehicles/Saurer2DM.h" // Swiss Saurer 2DM R6 Diesel truck
// Russian trucks --------
// #include "vehicles/Ural4320.h" // URAL 4320 6x6 V8 Diesel military truck
// #include "vehicles/Ural375D.h" // URAL 375D 6x6 V8 petrol military truck
// #include "vehicles/URAL375.h" // URAL 375D 6x6 V8 petrol military truck (new version with better V8 sound, but good bass speaker required)
// #include "vehicles/GAZ66.h" // GAZ-66 V8 petrol military truck
// Russian tanks -------
// #include "vehicles/IS3.h" // IS-3 WW2 battle tank, V12 Diesel (dual ESC mode, good bass speaker required)
// Tractors -------
// #include "vehicles/KirovetsK700.h" // Russian Kirovets K700 monster tractor. Extreme turbo sound!
// #include "vehicles/FergusonTEA20.h" // 1940ies Ferguson TEA20 petrol engine tractor
// Excavators -------
//#include "vehicles/Volvo EC550ElExcavator.h" // Volvo EC550EL 55 ton ecxavator (use "FLYSKY_FS_I6S_EXCAVATOR" remote profile and SERVOS_EXCAVATOR servo profile)
//#include "vehicles/Caterpillar323Excavator.h" // Caterpillar 323 excavator (use "FLYSKY_FS_I6S_EXCAVATOR" remote profile)
#include "vehicles/Pingon14CExcavator.h" // Pingon 14C excavator (use "FLYSKY_FS_I6S_EXCAVATOR" remote profile)
// Dumpers -------
// #include "vehicles/Benford3TonDumper.h" // Benford 3 ton dumper
// Wheel Loaders -------
// #include "vehicles/VolvoL120H.h"
//#include "vehicles/HitachiZW370.h" // Hitachi ZW 370-6 heavy wheel loader (for Lukas Cajkar Harmony 370 3D printe
Uncomment vehicle you want, ensure only one is uncommented
// Articulated Dump Trucks -------
// #include "vehicles/CAT730.h" // Caterpillar 730 heavy dump truck
// US motorcycles --------
// #include "vehicles/HarleyDavidsonFXSB.h" // Harley Davidson FXSB V2 motorcycle
// US cars --------
// #include "vehicles/ChevyNovaCoupeV8.h" // 1975 Chevy Nova Coupe V8
// #include "vehicles/1965FordMustangV8.h"// 1965 Ford Mustang V8
// #include "vehicles/Chevy468.h" // Chevy 468 big block V8
// EU cars --------
// #include "vehicles/VwBeetle.h" // VW Käfer / Beetle
// #include "vehicles/JaguarXJS.h" // Jaguar XJS V12, manual transmission
// #include "vehicles/JaguarXJSautomatic.h" // Jaguar XJS V12, automatic transmission
// #include "vehicles/MGBGtV8.h" // MGB GT V8, manual transmission
// #include "vehicles/LaFerrari.h" // Ferrari LaFerrari, V12, 6 speed double clutch transmission
Default (Rc_Engine_Sound_ESP32-test)
Auto
Ln 57, Col 113
Spaces: 4
UTF-8
LF
C++
Finish Setup
PlatformIO



Uncomment remote you want, ensure only one is uncommented. Scroll down for channel settings


Rc_Engine_Sound_ESP32-test
EXPLORER
RC_ENGINE_SOUND_ESP32...
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10_Trailer.h
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README.md
OUTLINE
1_Vehicle.h
3_ESC.h
src > 3_ESC.h > ...
// Additional reverse speed (disconnect & reconnect battery after changing this setting):
// - Meccano Dumper = 0
// - RZ7886 Driver = 0
uint16_t escReversePlus = 0;
// Crawler mode escRampTime (see "8_Sound.h") WARNING: a very low setting may damage your transmission!
uint8_t crawlerEscRampTime = 10; // about 10 (15 for Jeep), less = more direct control = less virtual inertia
// Allows to scale vehicle file dependent acceleration
uint16_t globalAccelerationPercentage = 100; // about 100 - 200% (200 for Jeep, 150 for 1/8 Landy) Experimental, may cause automatic transmission shifting issues!
// BATTERY PROTECTION SETTINGS **
/* Battery low discharge protection (only for boards with voltage divider resistors):
 * IMPORTANT: Enter used resistor values in Ohms (Ω) and THEN adjust DIODE_DROP, until your readings match the actual battery voltage! */
// #define BATTERY_PROTECTION // This will disable the ESC output, if the battery cutout voltage is reached. 2 fast flashes = battery error!
const float CUTOFF_VOLTAGE = 3.3; // Usually 3.3 V per LiPo cell. NEVER below 3.2 V!
const float FULLY_CHARGED_VOLTAGE = 4.2; // Usually 4.2 V per LiPo cell, NEVER above!
const float RECOVERY_HYSTERESIS = 0.2; // around 0.2 V
/* Note on resistor values: These values will be used to calculate the actual ratio between these two resistors
 * When selecting resistors, always use two of the same magnitude: Like, for example, 10k/2k, 20k/4k or 100k/20
 * WARNING: If the ratio is too LOW, like 10k/5k (2:1 = 2), the battery voltage will most likely DAMAGE the con
 * Example calculation: 2000 / (2000 + 10000) = 0.166 666 666 7; 7.4 V * 0.167 = 1.2358 V (of 3.3 V maximum on GPIO Pin). */
uint32_t RESISTOR_TO_BATTTERY_PLUS = 9400; // Value in Ohms (Ω), for example 10000
uint32_t RESISTOR_TO_GND = 1000; // Value in Ohms (Ω), for example 2000. Measuring exact resistor values before soldering, if possible is recommended!
float DIODE_DROP = 0.31; // Fine adjust measured value and/or consider diode voltage drop (about 0.34V for SS34 diode)
/* It is recommended to add a sticker to your ESP32, which includes the 3 calibration values above */
volatile int outOfFuelVolumePercentage = 80; // Adjust the message volume in %
// Select the out of fuel message you want:
#include "vehicles/sounds/OutOfFuelEnglish.h"
// #include "vehicles/sounds/OutOfFuelGerman.h"
// #include "vehicles/sounds/OutOfFuelFrench.h"
// #include "vehicles/sounds/OutOfFuelDutch.h"
// #include "vehicles/sounds/OutOfFuelSpanish.h"
// #include "vehicles/sounds/OutOfFuelPortuguese.h"
// #include "vehicles/sounds/OutOfFuelJapanese.h"
// #include "vehicles/sounds/OutOfFuelChinese.h"
// #include "vehicles/sounds/OutOfFuelTurkish.h"
// #include "vehicles/sounds/OutOfFuelRussian.h"
If Battery Protection resistors are not fitted, ensure this is commented out.
Default (Rc_Engine_Sound_ESP32-test)
Auto
Ln 135, Col 155
Spaces: 2
UTF-8
LF
C++
Finish Setup
PlatformIO

INTERNAL